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ZIXIN YIN, Beihang University, Beijing, China

JIAKAI WANG, Beihang University, Beijing, China

YIFU DING, Beihang University, Beijing, China

YISONG XIAO, Beihang University, Beijing, China

JUN GUO, Beihang University, Beijing, China

RENSHUAI TAO, Beihang University, Beijing, China

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Improving Generalization of Deepfake Detection with Domain Adaptive Batch Normalization

Zixin Yin
yzx835@buaa.edu.cn
Beihang University
Beijing, China

Jiakai Wang*
jk_buaa_scse@buaa.edu.cn
Beihang University
Beijing, China

Yifu Ding
zjdyf@buaa.edu.cn
Beihang University
Beijing, China

Yisong Xiao
18373494@buaa.edu.cn
Beihang University
Beijing, China

Jun Guo
junguo@buaa.edu.cn
Beihang University
Beijing, China

Renshuai Tao
rstao@buaa.edu.cn
Beihang University
Beijing, China

Haotong Qin
qinhaotong@buaa.edu.cn
Beihang University
Beijing, China

ABSTRACT

Deepfake, a well-known face forgery technique, has raised serious concerns about personal privacy and social media security. Therefore, a plenty of deepfake detection methods come out and achieve outstanding performance in the single dataset case. However, current deepfake detection methods fail to perform strong generalization ability in cross-dataset case due to the domain gap. To tackle this issue, we propose Domain Adaptive Batch Normalization (DABN) strategy to mitigate the domain distribution gap on different datasets. Specifically, DABN utilizes the distribution statistics of the testing dataset in replace of the original counterparts so as to avoid distribution mismatch and restore the effectiveness of BN layers. Equipped with our DABN, detection method can be more robust when generalized into a broader usage. Note that our method is flexible and can be further employed on most existing deepfake detection methods during testing, which shows a great practical value. Extensive experiments on multiple datasets and models demonstrate the effectiveness of DABN. The proposed method achieves an average accuracy improvement by nearly 20% of existing strategies on Celeb-DF dataset under black-box settings, indicating strong enhancement of generalization ability of the deepfake detection models.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision tasks**; • **Security and privacy** → *Social aspects of security and privacy.*

*Corresponding author

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KEYWORDS

Deepfake detection, domain adaptation, batch normalization, deep neural network

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1 INTRODUCTION

Deepfake, a new facial forgery technique, is becoming more prominent in social media and filming industry with the development of the deep generator. Variational Autoencoders [25] (VAE) and Generative Adversarial Networks [8] (GAN) are well-known deepfake methods, which is able to transpose one's face to another's, *i.e.* swap face in short. Since the abuse of deepfake, distinguishing the fake faces from real ones intrigues researchers to devote to deepfake detection methods. Current deepfake detection methods can be roughly categorized into two types, heuristic methods [1, 2, 6, 19, 23, 34, 35] and DNN-based methods [3, 9, 10, 18, 20, 22, 27]. Both have achieved outstanding performance in single dataset case. However, when it turns to cross-dataset cases, most existing deepfake detection methods might suffer a catastrophic performance drop, as shown in Figure 1. It limits the broader use of these methods.

We attribute this observation to domain gap, *i.e.*, the distribution differences between different domains, which has been long proposed in various fields [7, 12] but does not receive due attention in deepfake detection field. Early works focus more on deepfake generation process. Recent work [33] eliminates the domain gap by style transferring from the source domain to the target one, and indicates that the domain gap naturally exists in different datasets might be responsible for the poor generalization of detection methods. [33] provide a possible solution but requires to train a transferring network, which is kind of time-consuming.

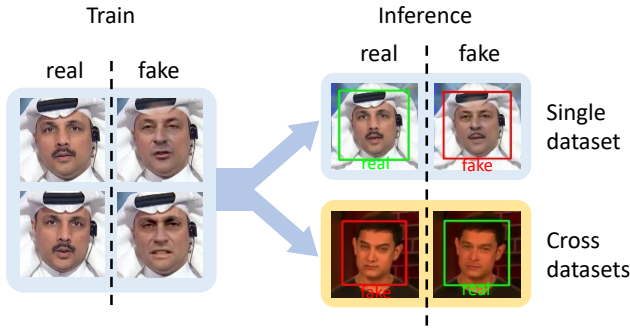


Figure 1: Detection methods usually perform well in single-dataset case (training and testing samples are from the same dataset), while fail to be generalized in cross-dataset case (training and testing samples are from different datasets).

In this paper, we first revisit the deepfake detection task from the perspective of distribution statistics of various datasets and provide more insights for the domain gap problem. We illustrate the domain gap by comparing the distribution statistics of several datasets. However, the original statistics in BN layer are derived from the training dataset, while the testing data may have quite different distribution statistics, so the original statistics in BN layer might not be appropriate for normalization in cross-dataset case. Therefore, it would cause severe performance drop if the inference network still uses the original statistics in Batch Normalization.

Motivated by the domain gap problem which makes the BN layer run into the risk of invalidation, we propose the Domain Adaptive Batch Normalization (DABN) to restore the effectiveness of BN layer, and further mitigate the domain gap of different datasets. Specifically, DABN utilizes the distribution statistics of the testing data to replace the original counterparts while performing normalization in BN layer. Therefore, the BN layer would be able to provide appropriately normalized data and deliver a reasonable distribution to the following layers, and eventually make better predictions. In this way, there is no need of an extra network that transfers the testing dataset to match the source one, but directly shift the distribution in inference process.

Extensive experiments on multiple deepfake detection dataset (e.g., Celeb-DF [21], FaceForensics++ [26], UADFV [19], DF-TIMIT [16]) utilizing several models (e.g., ResNet18 [11], Xception [5] and the state-of-the-art model, SPSL [22]) demonstrate that the models with DABN can improve the generalization of detection networks and boost their performance in cross-dataset cases. Models equipped with DABN outperform the vanilla ones by a large margin. The average accuracy improvement of ResNet18, Xception, SPSL models trained on Celeb-DF dataset while tested in others is up to 23.48%, 19.11% and 17.99% respectively. Especially when testing in UADFV dataset, DABN helps the ResNet18 to have a 31.37% accuracy gain. Our contributions can be listed as follows:

- We propose the Domain Adaptive Batch Normalization to mitigate the domain gap between training and testing dataset and further improve the deepfake detection methods to be generalized into unseen testing datasets.

- Extensive experiments demonstrate that models with Domain Adaptive Batch Normalization outperform the vanilla models by a large margin.
- Our DABN approach is flexible which could be easily employed on various deepfake detection methods, which shows a great practical value.

2 RELATED WORKS

In this section, we give a brief introduction about the current deepfake generation techniques and deepfake detection methods.

2.1 Deepfake Generation

Deepfake generation techniques can be roughly categorized into four classes [32]. 1) Face synthesis methods create entire non-existent face images, usually with the powerful GANs (e.g., StyleGAN [15], ProGAN [14]). 2) Face swap, also known as identity swap, replaces the face of one person with the others to swap the identity of the person, commonly carried out by classical computer graphics-based techniques (i.e., FaceSwap¹ (FS)), novel deep learning techniques (i.e., DeepFakes² (DF)), and GAN-based approaches (e.g., CycleGAN [36]). 3) Face editing can modify some attributes of the face, like the manipulation in FaceApp mobile application³ in which users can change the color of skin and hair, the age, the gender, etc. This manipulation process is usually generated through GANs such as the StarGAN [4]. 4) Face reenactment consists of modifying the facial expression. Two most popular techniques are Face2Face [26] (F2F) and NeuralTextures [31] (NT), which replace the facial expression of one person with the other one's. With the developing of deep learning, more deepfake generation methods come out, calling for deepfake detection methods with high generalization capability.

2.2 Deepfake Detection

In the past few years, many methods were proposed to distinguish the deepfake manipulated faces from the real ones. Existing deepfake detection methods can be roughly divided into two categories, heuristic methods [1, 2, 6, 19, 23, 34, 35] and DNN-based methods [3, 9, 10, 18, 20, 22, 27]. Heuristic-based methods use some known or hand-made heuristics to detect deepfake. Some methods utilize the craft machine learning features [35] or some relatively simple visual aspects [23] such as missing details in the eye and teeth areas, etc. Human biometrics (e.g., facial expressions [34], head movements [1] and eye blinking [6, 19]) are also used in deepfake detection.

Deep learning has shown outstanding performance in many areas [17, 24, 30], and also has a great deal of potential in deepfake detection. DNNs are utilized to detect artifacts introduced by the face manipulation pipeline. Y. Li and S. Lyu [20] use Convolution Neural Networks to detect the limited resolution and the artifacts in affine transforms. L. Li et al. [18] focus on the blending boundary of two faces in deepfake generation processes. Some DNN-based deepfake detection methods rely on temporal-level inconsistency, utilizing the combination of RNNs and CNNs [9, 27] or 3D CNNs

¹<https://github.com/MarekKowalski/FaceSwap>

²<https://github.com/deepfakes/faceswap>

³<https://www.faceapp.com/>

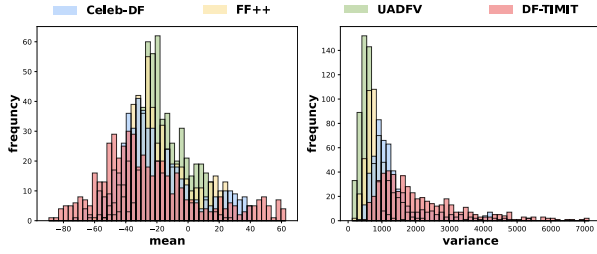


Figure 2: Distribution statistics (i.e., the mean and variance) of different datasets.

Table 1: Publicly available datasets with different conditions of data collection.

Dataset	Real Videos (Source)	Fake Videos (Techniques)
UADFV [19]	49 (Youtube)	49 (FakeApp)
DF-TIMIT [16]	620 (VidTIMIT [28])	620 (faceswap-GAN)
FaceForensics++ [26]	1000 (Youtube)	4000 (FS, DF, F2F [26], NT [31])
Celeb-DF [21]	890 (Youtube)	5639 (DF)

(e.g., I3D [3], 3D ResNet [10]). With usage of frequency features, especially phase spectrum, H. Liu et al. [22] propose SPSP method to detect the artifacts introduced in up-sampling processes, which achieves the state-of-the-art performance on cross-datasets evaluation. However, these methods focus on distinguishing the artificial facial aspects, but neglect the domain gap naturally exists under different conditions. Toward this end, we propose Domain Adaptive Batch Normalization, which concerns the domain gap problem, to further improve the generalization capability of existing detection networks.

3 METHOD

In this section, we first pay close attention to the distribution statistics over different datasets, and illustrate the domain gap between datasets generated by deepfake. To tackle this problem, we propose the Domain Adaptive Batch Normalization (DABN) method, which enables the model to exploit the distribution statistics of testing data to provide a reasonable Batch Normalization.

3.1 Analysis of the Domain Gap Problem

Domain gap has been investigated in various fields [7, 12]. A well-known explanation is that, the DNN models trained in source domain (i.e., training set) perform well in the domain with the same independent and identically distributed (i.i.d.) setting, but fail to be generalized to the domain with an out-of-distribution (o.o.d.) setting [29]. The cause could be superimposed, including the conditions and contents of different datasets, such as the light, resolution and so on.

Current DeepFake datasets consist of two parts. Real faces are from videos widely collected in the real-world, and DeepFake faces are created by various generation methods. For example, as shown in Table. 1, UADFV [19] contains 49 real videos from Youtube and 49 fake videos generated by the FakeApp mobile application, while FaceForensics++ [26] has 1000 real videos from Youtube and 4000

fake videos created using FaceSwap, DeepFakes, Face2Face [26], and NeuralTextures [31]. These deepfake datasets may have very different attributes and properties. We dub the phenomenon as domain gap.

We revisit the deepfake detection task from the perspective of distribution statistics. Figure 2 shows a close-up of the domain gap phenomenon between several well-known deepfake detection datasets, including Celeb-DF [21], FaceForensics++ [26], UADFV [19], DF-TIMIT [16]. We extract feature maps using the network and analyze the mean and variance in channels. As can be seen in Figure 2, distribution statistics vary from different datasets. For example, mean of DF-TIMIT dataset is in the range of $(-61, 125)$, while that of UADFV is $(-53, 26)$. Also, the variance of DF-TIMIT ranges from 491 to 7129, while UADFV is from 121 to 1612. The distribution discrepancies should be responsible for the poor generalization of existing detection methods in practical use.

3.2 Domain Adaptive Batch Normalization

Normalization is a typical method used to mitigate the negative impact of domain gap [29]. The commonly used normalization method in most DNN models is Batch Normalization (BN), which is proposed to scale and shift the value of network activations during training [13]. In general, the original BN module is fixed in inference phase, which means the network utilizes the original BN statistics to normalize the test data. However, distributions from different datasets can be quite different, which may result in the invalidity of performing normalization. We give a detailed explanation in a statistical way.

Considering a well-trained DNN detection model. Assume that the density of source distribution is $p_s(y|x) : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$, and that of target distribution is $p_t(y|x) : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$, where $\mathcal{X}$ denotes the input data and $\mathcal{Y}$ is the corresponding labels. Since the parameters of the model is frozen, we can get $p_s(y|x) = p_t(y|x)$. [29] leverages the probability $p_s(x, y) = p_s(x)p_s(y|x)$ and $p_t(x, y) = p_t(x)p_t(y|x)$ to define the covariate shift. In cross-dataset case, $p_s(x)$ and $p_t(x)$ are naturally unequal due to the domain gap, therefore $p_s(x, y) \neq p_t(x, y)$. It can also be written as follow with vanilla BN:

$$p_s(y|x)p_s\left(\frac{x - \mathbb{E}_s(x)}{\sqrt{\mathbb{V}_s(x)}}\right) \neq p_t(y|x)p_t\left(\frac{x - \mathbb{E}_s(x)}{\sqrt{\mathbb{V}_s(x)}}\right) \quad (1)$$

We propose Domain Adaptive Batch Normalization (DABN) to remove the covariate shift by performing normalization to $p_t(x)$, aiming to eliminate the distribution discrepancy between the normalized $p_t(x)$ and $p_s(x)$. Therefore, we replace the original BN layer statistics with the testing dataset statistics while performing normalization. Eqn. 2 can be rewritten as:

$$p_s(y|x)p\left(\frac{x - \mathbb{E}_s(x)}{\sqrt{\mathbb{V}_s(x)}}\right) \approx p_t(y|x)p\left(\frac{x - \mathbb{E}_t(x)}{\sqrt{\mathbb{V}_t(x)}}\right) \quad (2)$$

In Eqn. 2, $\mathbb{E}_s(\cdot)$ and $\mathbb{V}_s(\cdot)$ represent the expectation and variance in source domain, $\mathbb{E}_t(\cdot)$ and $\mathbb{V}_t(\cdot)$ represent the expectation and variance in target domain. In this way, DABN normalizes the data into the standardized normal distribution, which is similar to the distribution during training.

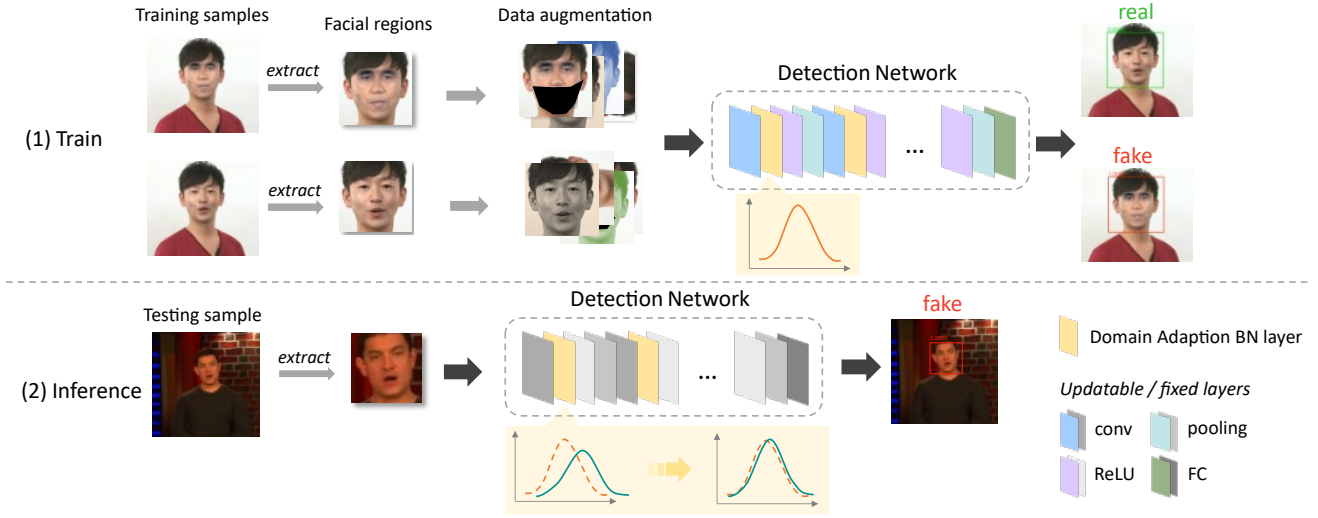


Figure 3: The overall deepfake detection pipeline in training process and inference process. We have the detection network trained on training dataset. While in the inference process, we replace the traditional BN with the Domain Adaptive Batch Normalization (DABN) to restore the effectiveness of normalization, while keep other layers fixed. DABN makes the normalization adaptive to the testing dataset and delivers a reasonable distribution to the following layers. It mitigates the domain gap between different datasets and eventually helps the model to be generalized into unseen data.

During inference, we compute the mean and variance as $\mathbb{E}_t$ and $\mathbb{V}_t$ in Eqn. 3 and Eqn. 4 with the following equations:

$$\mu_B = \alpha \cdot \mu_B + (1 - \alpha) \cdot \frac{1}{M} \sum_{i=1}^M \mathbf{x}_i \quad (3)$$

$$\sigma_B^2 = \alpha \cdot \sigma_B^2 + (1 - \alpha) \cdot \frac{1}{M} \sum_{i=1}^M (\mathbf{x}_i - \mu_B)^2 \quad (4)$$

where μ_B denotes the mean value, σ_B^2 denotes the variance value, $\mathbf{x}_i$ is the input vector of mini-batch with size of M , and α denotes the momentum parameter (usually set to 0.1).

In a nutshell, DABN restores the effectiveness of BN layer by normalizing the data using the testing distribution statistics, avoiding the distribution mismatch caused by domain gap. Equipped with DABN, the model is able to provide appropriately normalized data and deliver a reasonable distribution to the following layers. Because the expectations and variance towards the target domain ($\mathbb{E}_t(\cdot)$ and $\mathbb{V}_t(\cdot)$) only depend on the dataset, therefore it is easy to be generalized in cross-dataset case.

3.3 Overall Deepfake Detection Process

The overall deepfake detection process pipeline is shown in Figure 3, including three main parts, including data processing, feature extraction, classification.

In the data processing phase, we first extract deepfake manipulated images from deepfake videos. And we use Face detector and facial key-point extractor to extract facial regions. After data processing, we can get the input images $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ and corresponding labels $\mathcal{Y} = \{y_1, \dots, y_N\}$.

In the training phase, the images $\mathbf{x}_i \in \mathcal{X}$ processed by data augmentation $t \in \mathcal{T}$ are inputted into DNN model $\mathcal{F}$. $\hat{y}_i = \mathcal{F}(t(\mathbf{x}_i))$

is the prediction of the model $\mathcal{F}$ for given $\mathbf{x}_i$. We optimize the network by minimizing the cross entropy loss $\mathcal{L}$ deliver in Eqn. 5 during training.

$$\mathcal{L} = \text{CrossEntropy}(\hat{y}_i, y_i) \quad (5)$$

As shown in Algorithm 1, in inference process, we apply the DABN to replace the traditional BN in DNN model to get the new model $\mathcal{F}'$. And then we get the prediction results $\hat{y}_i = \mathcal{F}'(\mathbf{x}_i)$ for the given test images $\mathbf{x}_i \in \mathcal{X}$.

Algorithm 1 Inference Process of DeepFake Detection

Input: trained DNN model $\mathcal{F}$, test set $\mathcal{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$

Output: prediction results $\hat{\mathcal{Y}} = \{\hat{y}_1, \dots, \hat{y}_N\}$

replace the BN layers with the DABN module and get $\mathcal{F}'$
process the input data $\mathcal{X}$, extract facial region

for each $\mathbf{x}_i \in \mathcal{X}$ **do**

$\hat{y}_i \leftarrow \mathcal{F}'(\mathbf{x}_i)$ // learn the distribution while testing

end for

Our DABN brings little overhead for the detection network with no need of retraining or finetuning. It is remarkable that the DABN method is flexible and thus can be employed on most existing deepfake detection methods during testing, which shows a great practical value.

4 EXPERIMENT

In this section, we conduct lots of experiments over different datasets and models to verify the effectiveness of Domain Adaptive Batch Normalization (DABN) by comparing the model equipped with DABN against the vanilla ones. We evaluate our DABN in single-dataset and cross-dataset cases, and demonstrate that our DABN

can help to improve the performance in both. Finally, we also visualize the distribution statistics about the feature maps with/without our DABN, and possible explanations about the effectiveness of DABN is discussed.

4.1 Experimental Settings

Evaluation metrics. Following [21, 26], the Accuracy (ACC) and the Area Under ROC Curve (AUC) metrics are used to evaluate the detection ability and generalization ability, which are commonly used in deepfake detection field.

Network Architecture. To evaluate the effectiveness and versatility of our method, we integrate the DABN into various commonly used models, including ResNet18 [11], Xception [5], and the state-of-the-art SPSSL [22] model. We compare the performance of the models equipped with our method against the original models.

Datasets. We train the models on two public deepfake datasets: Celeb-DF [21] and FaceForensics++(FF++) [26], while test the models on four public deepfake datasets: Celeb-DF [21], FF++ [26], UADFV [19], and DF-TIMIT [16].

Implementation details. We extract every ten frames of the videos in the datasets and process the frames with MTCNN to extract the facial region. We randomly copy the real images to balance the quantity of positive and negative samples, and train the DNN models on the Celeb-DF and FF++ datasets for 50 epochs with data augmentations⁴. We adopt Adam optimizer and the learning rate is initialized to 0.003, which is decayed by 0.3 at the 3rd, 8th, 15th, 25th, and 40th epoch. We conduct the training and testing processes on an NVIDIA Tesla V100-SXM2-16GB GPU. The batch size is 512 for ResNet18 [11], 96 for Xception [5] and SPSSL [22].

Table 2: The detection ability of deepfake detection methods in single dataset, which means the training and testing samples are from the same dataset.

Model	Celeb-DF		FF++	
	ACC(%)	AUC(%)	ACC(%)	AUC(%)
ResNet18	90.10	99.45	81.79	90.42
+DABN	95.81	99.54	85.79	91.32
Xception	98.41	99.80	87.70	92.27
+DABN	96.97	99.79	88.10	92.36
SPSSL	96.52	99.76	87.98	92.43
+DABN	96.80	99.83	87.94	92.30

4.2 Single Dataset Detection Ability

We evaluate the performance in detection ability of our DABN in the single dataset case. Specifically, we train ResNet18, Xception and SPSSL models with and without the DABN modules in Celeb-DF and FF++ datasets, and test the models in the same datasets to evaluate the detection ability. Table 2 shows that current deepfake detection methods perform strong detection ability when training and testing samples are from the same dataset. In the meanwhile, our DABN method can further improve the detection performance

of DNN models, which helps ResNet18 to have up to 5.71% increase in Celeb-DF dataset, and 4.0% in FF++ dataset.

4.3 Cross Datasets Generalization Ability

In this work, we evaluate the performance in generalization ability of our DABN in cross-datasets case, which means the training and testing samples are from different datasets. We train the ResNet18, Xception and SPSSL models with and without DABN modules in Celeb-DF and FF++ datasets, while test the models in other datasets (e.g., UADFV, DF-TIMIT) to evaluate the generalization ability. Table 3 and Table 4 show the models trained with Celeb-DF and FF++ respectively, while tested with other datasets. We can draw conclusions from the results that 1) Current deepfake detection methods are suffering from the catastrophic performance drop in the generalization ability, e.g., the accuracy of ResNet18 trained in Celeb-DF dataset drop to 50.51% in FF++ dataset, compared with 90.10% in Celeb-DF dataset as mentioned in Sec. 4.2. 2) DABN can boost the generalization ability of the model in cross-dataset cases. The average accuracy improvements of DABN in ResNet18, Xception, SPSSL models are 23.48%, 19.11%, 17.99% respectively when trained in Celeb-DF dataset (see Table 3), and 6.55%, 1.19%, 4.50% when trained in FF++ dataset (see Table 4). And it is notable that the improvement of ResNet18 tested in UADFV dataset is up to 31.37% and in DF-TIMIT dataset is 31.05%. Comparing with the vanilla models, DABN revives the detection network to be effective in cross-dataset cases regardless of the domain gap between the training and testing datasets.

Table 3: The generalization ability of deepfake detection methods, which are trained in Celeb-DF dataset and tested in FF++, UADFV and DF-TIMIT datasets.

Model	FF++		UADFV		DF-TIMIT	
	ACC(%)	AUC(%)	ACC(%)	AUC(%)	ACC(%)	AUC(%)
ResNet18	50.51	64.09	52.66	86.53	51.76	87.12
+DABN	58.53	62.23	84.03	92.61	82.81	89.75
Xception	51.26	63.18	62.89	93.05	53.12	90.54
+DABN	59.44	64.47	88.24	97.12	76.93	88.84
SPSSL	51.37	65.41	55.89	83.60	62.96	96.67
+DABN	60.62	64.39	85.39	94.92	78.17	83.69

Table 4: The generalization ability of deepfake detection methods, which are trained in FF++ dataset and tested in Celeb-DF, UADFV and DF-TIMIT datasets.

Model	Celeb-DF		UADFV		DF-TIMIT	
	ACC(%)	AUC(%)	ACC(%)	AUC(%)	ACC(%)	AUC(%)
ResNet18	52.45	66.06	73.25	84.09	68.65	75.07
+DABN	62.53	67.78	74.23	83.47	77.24	86.01
Xception	61.13	74.33	84.17	92.12	86.21	94.25
+DABN	63.39	71.18	84.87	91.41	86.83	94.56
SPSSL	61.20	66.68	73.19	80.79	83.74	92.03
+DABN	64.17	71.93	82.13	89.07	85.34	91.73

⁴https://github.com/selimsef/dfdc_deepfake_challenge

4.4 Visualization about Domain Gap Mitigation

We also demonstrate the ability of DABN about mitigating the domain gap over different datasets. We train the ResNet18 model with the Celeb-DF dataset, and observe the distribution statistics in the feature map during inference using both the Celeb-DF and FF++ datasets to perform single-dataset and cross-dataset cases. Figure 4 shows the close-ups. The feature distribution of FF++ dataset with the traditional BN has an obvious shift compared with that of Celeb-DF, which is the training dataset. Fortunately, equipped with DABN, the distribution statistics of FF++ seem more close to the training dataset. It forcefully explains the better performance of our method: DABN restores the effectiveness of BN, which mitigates the domain gap between different datasets, delivers a reasonable distribution to the following layers, and further improves the generalization ability of models.

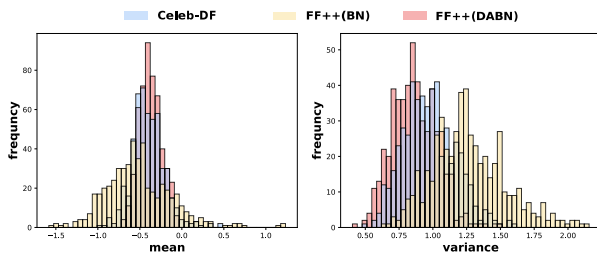


Figure 4: Distribution statistics of Celeb-DF and FF++(BN) datasets after performing DABN (i.e., FF++(DABN)). DABN can mitigate the domain gap between Celeb-DF dataset and FF++ dataset.

5 CONCLUSIONS

In our paper, we revisit the deepfake detection task from the perspective of distribution statistics of various datasets and analyze the domain gap problem in deepfake detection, which causes the distribution shift problem and makes the BN layer run into the risk of invalidation. Therefore, we propose the Domain Adaptive Batch Normalization (DABN) to restore the effectiveness of BN layer, and further mitigate the domain gap of different datasets. Specifically, DABN utilizes the distribution statistics of the testing data to replace the original counterparts while performing normalization in BN layer, which enables a reasonable normalization. Our DABN approach is flexible and could be easily employed on various deepfake detection methods, which shows a great practical value. Extensive experiments on multiple deepfake datasets and models demonstrate that models equipped with DABN outperform the vanilla ones by a wide margin, which is up to 23.48%, 19.11% and 17.99% for ResNet18, Xception, SPSSL models respectively. DABN is designed from the point of view of distribution statistics in data normalization, with no need of training a style transfer network to modify the testing images. Therefore, it greatly boosts the generalization ability of detection networks.

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